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EC4050
Literature Review

Game Theoretic Analysis of Euchre Using Probabilistic Methods

Introduction

Card games involving randomness are notoriously difficult to analyze. Natural features such as imperfect information, differing strategies, and the sheer number of combinations all add complexity. Euchre provides a keen example of this. Therefore, it becomes necessary to put constraints on different aspects of the game and make assumptions. The model derived here will focus on the decision of the dealer on whether to pick up a flipped-up Jack in the case where the dealer is left with a hand of all 9s and a 10 of the same suit as the Jack. Further it will be in the variation "Stick the Dealer" where the dealer will be forced to call a suit if everyone else passes. This creates a situation where the model relies on assuming each other player will make the best moves based on the expected return which is a basic assumption of strategic decision making. This rollback will then allow the dealer to determine if the risk reduction of picking up the Jack or the potential of someone else calling will result in a better expected payoff. The model in this paper attempts to answer this question using probability distributions to model the imperfect information that would be experienced in a game of euchre. It also uses these probability distributions to model the large number of combinations simplifying analysis. The result provides a model giving insight into the expected decisions that each player would make. It also determines that in the specific situation where the dealer is dealt a poor hand of 9s and 10s with a flipped Jack the best move would be to pass and not pick up the Jack.

Related literature

The existing literature discussing game theoretic models around card games has generally been toward solving them using combinatoric algorithms. Kirkpatrick (1954) describes a simple such model that evolves over the course of a game of "concentration" where both players attempt to match values of cards that are played face down. This model describes the way that assuming each player has perfect memory scoring will be expected to occur. This is further generalized in Demaine et al. (2001) which describes combinatorial games where all players have complete information, as well as describes the difficulty in determining ideal play for complex games. This work is generally limited though to perfect information games where players have full information whereas the model of Euchre outlined in this paper will need to be able to consider the fact that this is not the actual situation, and players will have access to asymmetric information.

The asymmetric information present requires players to consider different probability models. Probabilistic models similar to those that exist in this model are considered by Gorelik and Zolotova (2023) who built a model based on the ideas of minimizing variance of payoffs and maximizing payoffs. This model is limited, however, without the consideration of it being a zero-sum game where one player's gain is another's loss, and as such decision making has no competing interests. This is juxtaposed to the model of poker set forth by Sandholm (2010) who describes similar incomplete information games, where all players are in complete opposition as would exist in poker. Both models work with probabilities to determine payoff tables in sequential games; however, they differ in that the first doesn't assume opposition in the game and the second assumes that everyone is in total opposition. This is opposed to a game of euchre where there are four parties all of which have limited information and in which there is both cooperation and opposition.

In invoking poker, however, Sandholm (2010) does successfully tap into another important aspect of the game of euchre, such that one player's gain is another player's loss. The game of euchre is different in that both teams are attempting to race for points versus poker where players are competing for a limited set of chips. However, the players in opposition are mazimizing their own payoff and minimizing others is a common thread. This is connected to Cox, Jr. (2009) which describes risk analysis using game theory. In this work, a player in an attacker/defender relationship may allocate resources and a attacker may exploit this allocation. This describes quite succinctly the zero-sum nature of both poker and euchre where players may modify their behaviors and actions to minimize risks and determine where attackers may exploit them, and attackers may interpret these moves to their own benefit.

This model then relies this imperfect information and an asymmetrical information, both of which can lead to difficulty in finding a Nash equilibrium (Cox, Jr. 2009). These are concepts explored by Eguia et al. (2018) and Tuyls et al. (2018) who investigate effective analysis for such models. Eguia (2018) describes methods of simplifying solutions to imperfect information models if instead they are considered as symmetric information models. This is the basis of Tuyls (2018) which describes the ways in which an asymmetric game may be solved by using a symmetric counterpart. Using these together the model may be simplified to find an effective solution.

The model outlined in this paper will extend the contents of these previous papers by using combinatorial logic such as those outlined by Kirkpatrick (1954) and Demaine et al. (2001) to describe the probability tables similar to those used by Gorelik and Zolotova (2023) and Sandholm (2010). From here the risk analysis of Cox, Jr. (2009) can act as the basis for the overarching question of if the dealer should decide to pick up the Jack as they are in a defender position and attempting to prevent the unfortunate situation of being euchred. Then the work done by Tuyles et al. (2018) and Eguia et al. (2018) may be used as methodologies for simplifying and eventually solving this model.

Model

The model is a sequentially played game with imperfect information. Meaning each player makes decisions without complete knowledge of other players payoffs. The playthrough of this can be seen in Figure 1 below represented by a game tree. Where play starts with the Dealer, moves to Opponent 1, Teammate, Opponent 2. Then P1-P5 are the possible payoffs of the model. These payoffs, however, are probabilistic and each players individual knowledge will affect the actual selection. Specifically, due to the actual cards that each player receives when playing the game.

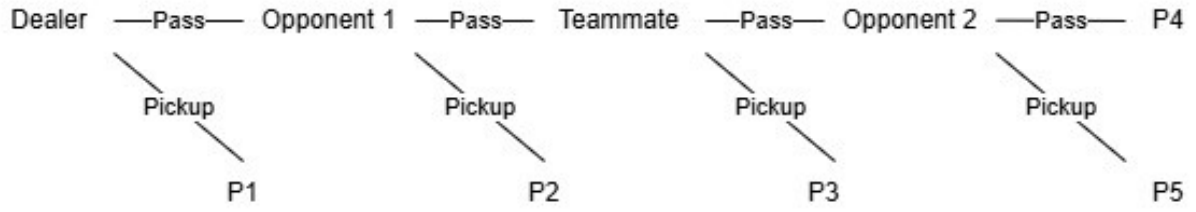

Figure 1. Generic Sequential Model of Play.

One of the natural difficulties that encounters a model of a card game with many players is the extreme number of combinations possible for assignments of the cards. Table 1 below shows the 24 different cards where each tile is assigned a random player, and where each player will have exactly 5 tiles, and the kitty will have exactly 4 tiles.

	9	10	J	Q	K	A
Spades	Dealer	Dealer	Dealer/Kitty			
Clubs	Dealer					
Hearts	Dealer					
Diamonds	Dealer					

Table 1. Shows the randomly assigned tiles to each player

It may also be noted the Dealer player automatically is assigned a set of tiles in the constrained model attempting to solve the question if the Dealer should pass on the Jack or pick it up. This information is not, however, known to the other players. Attempting to apply combination to the model then recognizes there are three players who will each receive exactly five cards in any order and kitty which will receive three cards in any order. Equation 1 then solves this combination.

$$\frac{18!}{5! * 5! * 5! * 3!} = 617{,}512{,}896$$

Equation 1. The combinations of randomly assigned cards.

Where the 18 factorial represents there are 18! ways in which to assign the cards, the 5! is for each of the players who could receive the cards in any order, and the 3! is for the kitty which may receive the cards in any order.

There are 617,512,896 unique combinations of the dealings of the cards even with the applied constraints it becomes extremely difficult to analyze every combination to determine the perfect strategy. Instead, this model utilizes probabilistic distributions and expected values to determine the best option each player should make to get the best payoff. This probabilistic method requires further assumptions about the game; however, it allows for more generalizable payoff equations and a simpler solution.

There are two primary probability distributions of interest in this model the first being the distribution of cards, which has been more generalized to "hand strength" on a scale from 0 to 1 where a higher value implies a greater likelihood of winning more tricks. An appropriate distribution would be Equation 2.

$$X_h \sim \text{Beta}(4, 4)$$

Equation 2. The distribution of strength of hands.

Where X_h is the distribution of hand strength. This simplifies the actual hands individuals are getting from the specific combinations of cards to the strength of the hand if provided with the opportunity to call a specific suit. Implying each individual's hand is completely independent and every individual could have a weak or a strong hand. This simplification, however, allows for a generally accurate distribution of hand strengths for a dealt hand with a mean hand strength of 0.5. This distribution can be seen in Figure 2 below.

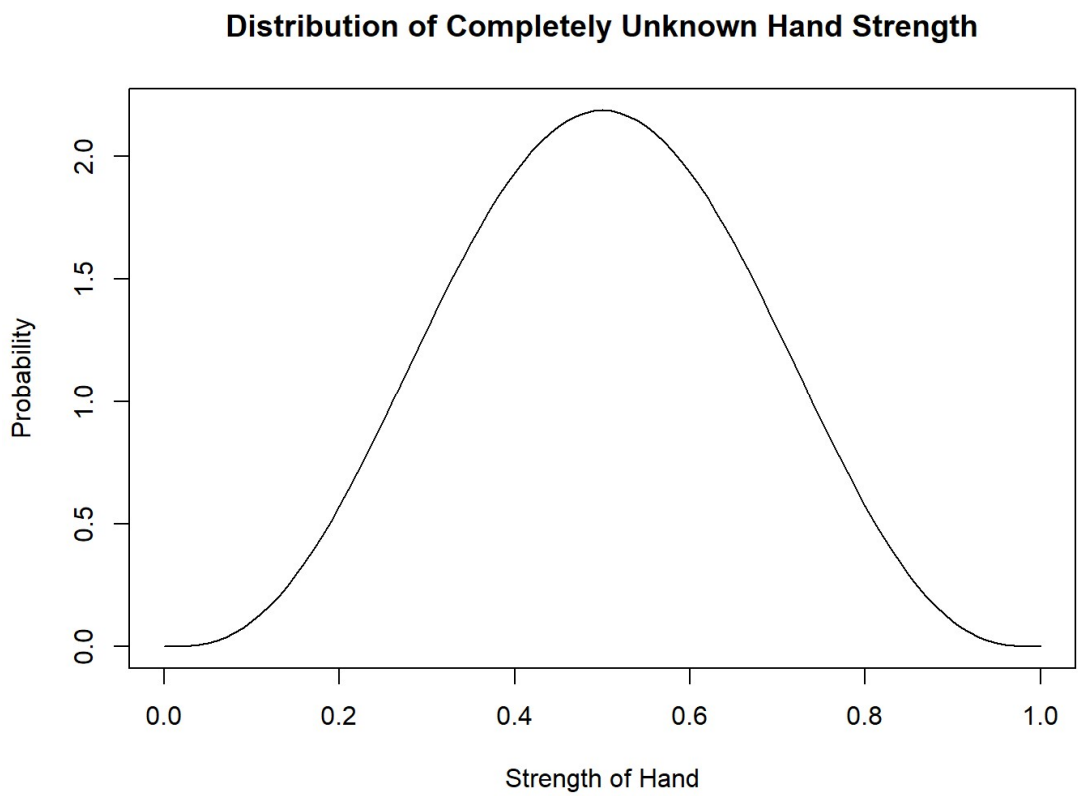

Figure 2. The distribution of a random players hand strength

Each player's hand strength still contains some variability in the number of tricks they are able to win. This distribution, however, is highly dependent on the strength of an individual's hand. Therefore the distribution described by Equation 3 is appropriate as it will change in response to x_h with the number of tricks expected to be taken increasing as x_h increases.

$$X_w|X_h \sim \text{BetaBin}(5, \text{alpha}, \text{beta})$$

$$\text{where alpha} = 5 * X_h \text{ and beta} = 5 - 5*X_h$$

Equation 3. The conditional distribution of winning hands depending on hand strength.

Where X_w is the distribution of won tricks and $X_w|X_h$ is the distribution of won tricks given the strength of the hand. This may be seen visually below in Figure 3 which describes the distributions of the number of tricks won given a specified x_h.

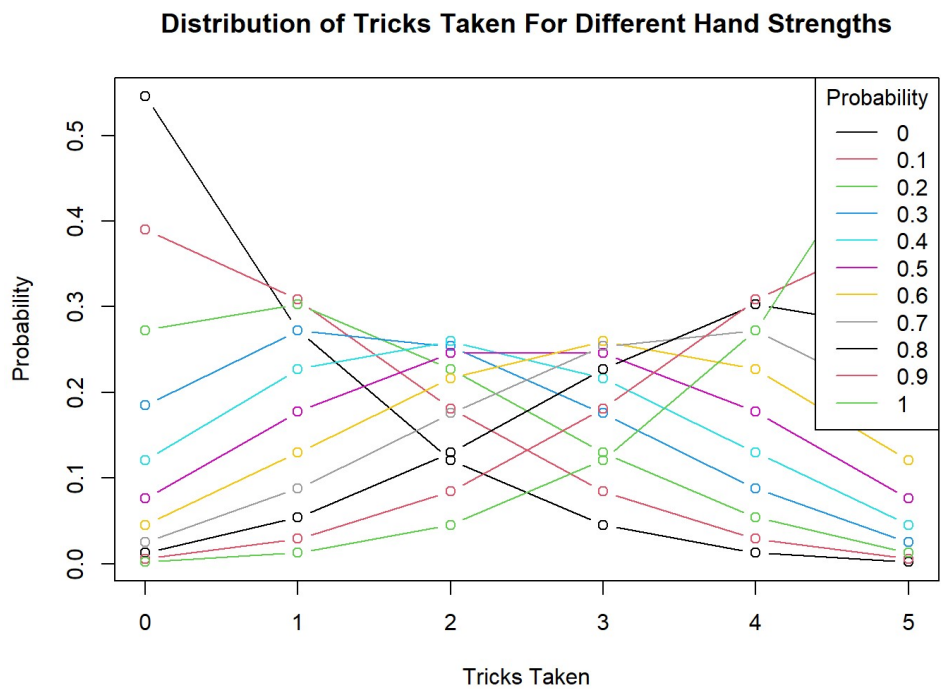

Figure 3. The conditional distributions of the number of tricks won depending on the strength of a hand.

These two distributions provide the tools to assess the hands and the number of tricks a individual will take given a specific hand and the distribution of hands. Where x_h is known to the player and therefore $P(X_w|X_h=x_h)$ can therefore be calculated to determine the expected number of tricks. However, the last needed tool is converting the number of tricks to payoffs. The following two piecewise functions provide the tools to determine the payoff and expected value for each player. It should be noted the Dealer and Teammate have equal payoffs as do Opponent 1 and Opponent 2 and each "teams" gain is the other team's loss. As the game is played to 10 tricks and therefore the more points the other team brings the former team closer to a loss. These piecewise functions converting number of tricks can be seen in Equation 4 which describes the payoff if the Dealer or Teammate calls it or in Equation 5 which describes the payoff if Opponent 1 or Opponent 2 calls the suit.

$$\begin{cases} (1,-1,1,-1) & 2 < X_w < 5 \\ (2,-2,2,-2) & X_w = 5 \\ (-2,2,-2,2) & X_w \leq 3 \end{cases}$$

Equation 4. The payoff table derived from the win distribution if Dealer or Teammate calls suit.

$$\begin{cases} (-1,1,-1,1) & 2 < X_w < 5 \\ (-2,2,-2,2) & X_w = 5 \\ (2,-2,2,-2) & X_w \leq 3 \end{cases}$$

Equation 5. The payoff table if Opponent 1 or Opponent 2 calls the suit.

At this point it is possible to begin to determine the rollback equilibrium on the decisions made along the sequential game tree.
Returning to Figure 1. The last decision made in the game tree is made by Opponent 2 who must decide between calling a suit and passing. Opponent 2 will make this decision based on consideration of the expected payoff if a pass is chosen. As Opponent 2 doesn't know the hand that the dealer has they must make an assessment based on a completely unknown hand strength as pictured in Fig 2. However, since the Dealer will be forced to call a suit regardless of hand strength the expected payoff can be determined from $P(X_w)$ of the dealer calling the suit. The probability distribution $P(X_w)$ of the Dealer as understood by Opponent 2 can be calculated by integrating over $P(X_w|X_h=x_h)$ for all x_h. The resultant probability distribution can be seen below in Figure 4.

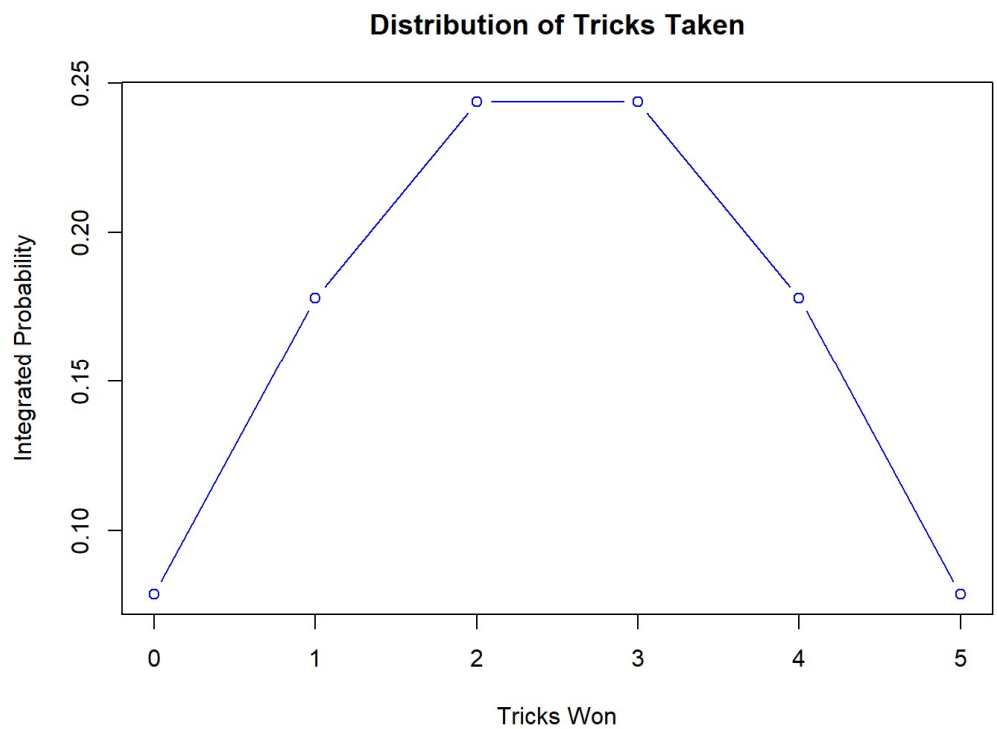

Figure 4. The distribution of tricks taken.

Taking this one step further and calculating the expected payoff by multiplying the probability by the value and summing each term this results in an expected payoff of (-0.366, 0.366, -0.366, 0.366). With this established as a baseline for what Opponent 2 believes the expected payoff for P4 to be it can also be established that Opponent 2 should choose to call if it would be expected that x_h would give a better expected payoff. This occurs when the hand strength is approximately 0.73 or higher which is a situation that happens in 10.2% of hands. By then integrating over the hands of equal strength or higher to find the average return of all of those hands returns a payoff of (-0.647, 0.647, -0.647, 0.647). Note that this is the average of only the hands that would be better than or equal to a hand strength of 0.73. These calculated values may then be applied to the game tree as the probability of the choices Opponent 2 will make and the expected return values, as seen in Figure 5 below.

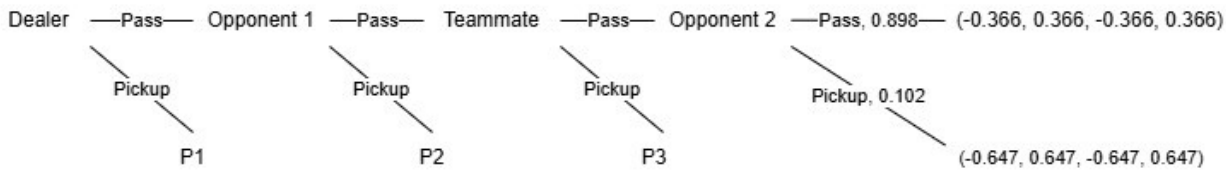

Figure 5. Game tree with calculations for Opponent 2

The expected outcome of passing from Teammate is then the two expected return values from Opponent 2 multiplied by the probability that they should choose each path. This can then be used as a baseline for what strength of cards Teammate should choose to pickup and the relative expected payoff. This may then be done again for Opponent 1. The resultant filled out decision tree is pictured below in Figure 6.

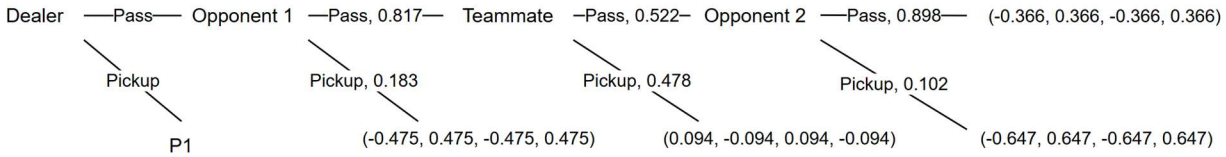

Figure 6. Filled out game tree from perspective all Opponent 1, Teammate, Opponent 2

Performing a rollback equilibrium on the Dealer, however, requires special consideration due to the nature of the outlined model. Where the Dealers hand has been defined to better describe the difference between being able to call initially for more favorable conditions and risk reduction. This will not affect the above model for each of the other players in the game as they don't know it, when the Dealer makes the decision, they have different knowledge of the payoffs if Opponent 2 passes and if they pickup the strength of their hand is set. To calculate the expected payoffs of these two decisions these different hands must be assigned strength. These have been assigned a value of 0.45 if called initially and 0.05 if not called and the Dealer is forced to call at the end of the game tree. The distributions of the expected number of tricks in for each of these situations are shown in Figure 7 and Figure 8 respectively.

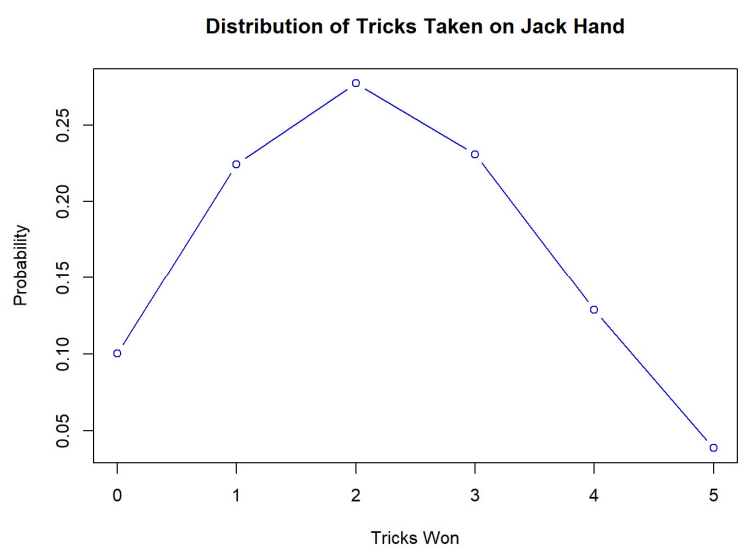

Figure 7. The distribution of the number of tricks taken if Dealer calls initially.

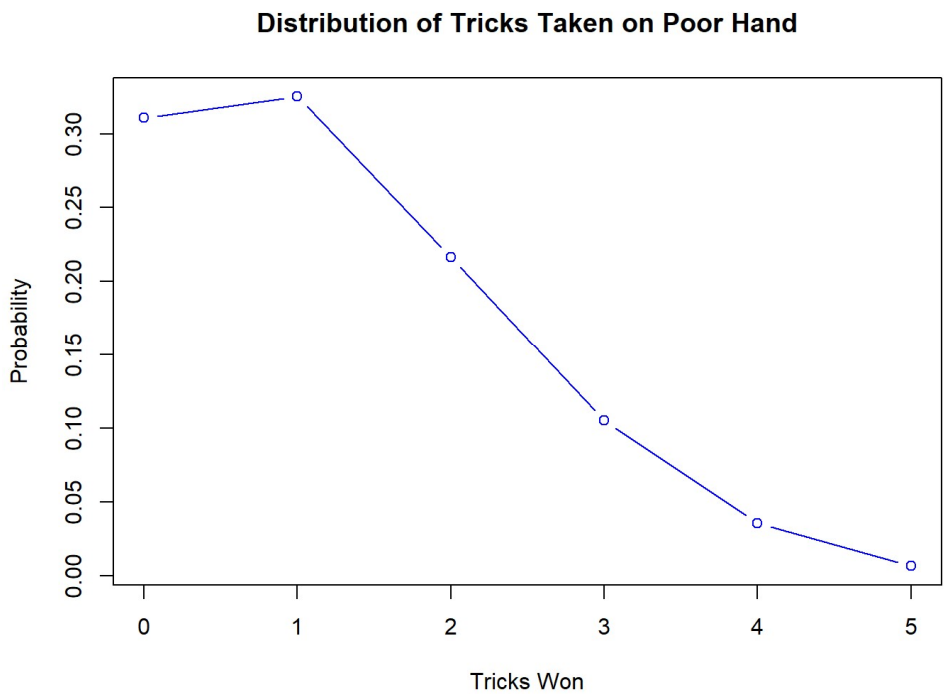

Figure 8. The distribution of the number of tricks the Dealer will take if calls at the end.

Using the known distribution in Figure 7 the payoffs at leaf P4 from Figure 1 can be recalculated, however, since none of the other players have this knowledge, their decisions will not be changed and therefore each of the probabilities will stay the same. Then using the known distribution in Figure 8 the expected payoff if the Dealer calls off the bat can be calculated. With these distributions in mind the following game tree in Figure 9 is the result.

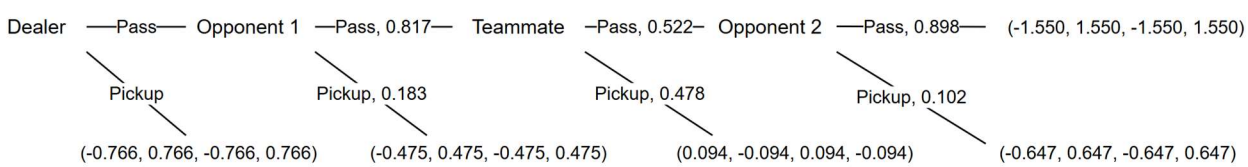

Figure 9. The game tree from the perspective of the dealer.

From the perspective of the Dealer the best move in this situation and the rollback equilibrium will rely on a expected value analysis depending on the expected moves of each of the following players. This assumes that each player will always play to their best strategic move maximizing their expected payoff which is required in strategic game theory. To determine the best choice for the Dealer then requires calculating the expected return of each decision. Picking up has an expected payoff of -0.766 and passing has an expected payoff of -0.672 indicating that in this model representing the game the Dealer would expect a better payoff by passing rather than picking up and therefore should choose to do so.

Conclusion

The derived model simplifies analysis of imperfect information using probability distributions to describe asymmetrical information. The results of the model imply the dealer and their teammate

have a significant disadvantage in this situation where either choosing to pick up the Jack or passing both have a negative payoff compared to Opponent 1 and Opponent 2 which have an expected positive payoff regardless of the decision made. The approximations and assumptions made also allow for new insights into the choices that should be made particularly for the other players in the game who are not considering constraints on the cards assigned to the Dealer. That the distribution of taken tricks is conditional upon the strength of a dealt hand and the specific strength of hand needed to call depends heavily on whether forcing the dealer to call will result in a “euche”. The model also demonstrates the idea of each player in the game tree being forced to determine their decision based only on the information provided to them. This is particularly demonstrated by each other player, making strategic decisions based on the idea that the Dealer’s hand is truly random even such as in the case where the Dealer has a really poor hand, but even though this is constrained in the model to answer the question the other players may not make use of this information and must play honestly based on their imperfect information.

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